

Advanced Programming Practices

MTech 2026

Lab - 5

General Instructions:

- Write one .c file per problem in the assignment, renamed to RollNo_APP5_Qn.c (e.g. ABC2026001_APP5_Q1.c).
 - Every program must compile without warnings using the gcc compiler.
 - Free any memory you allocate. Programs that leak memory or access out-of-bounds indices will lose marks even if the visible output looks correct.
 - Test each program with at least one normal case, one edge case (e.g. size 0 or 1, all-equal elements), and one larger randomly generated case.
 - Submit all the .c files on Google Classroom before the due time, any late submissions will be penalized or discarded.
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1. Write a C program to remove all duplicate elements from an integer array without using another array.
2. Write a C program to find the longest consecutive sequence in an array of integers.
3. Write a C program to separate an array into two arrays containing even and odd elements, respectively.
4. Write a C program to find all elements that are maximum in their row and minimum in their column.
5. Write a C program to count the number of vowels and consonants using recursion.